

CO-3 : ASSESSMENT TOOL – 1

CASE STUDY

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CASE STUDY 1 – EMPLOYEE SALARIES

PROBLEM IDENTIFICATION AND ANALYSIS

A company records the annual salaries (in ₹ lakhs) of **9 employees** as follows:

4, 5, 5, 6, 7, 8, 8, 9, 30

Using the above data,

1. Find the Mean.
2. Find the Median.
3. Find the Mode.
4. Find the Range.
5. Calculate the Population Variance.
6. Calculate the Population Standard Deviation.
7. Identify the Outlier.
8. Explain which measure of central tendency best represents the typical salary.

Given Data

Employee	Salary (₹ Lakhs)
1	4
2	5
3	5
4	6
5	7

Employee	Salary (₹ Lakhs)
6	8
7	8
8	9
9	30

Number of observations : $n = 9$ Theory

Mean

The mean is the average value of all observations.

It is calculated by dividing the total of all values by the number of observations.

Median

The median is the middle value after arranging the data in ascending order.

Mode

The mode is the value that occurs most frequently.

Range

The range is the difference between the maximum and minimum values.

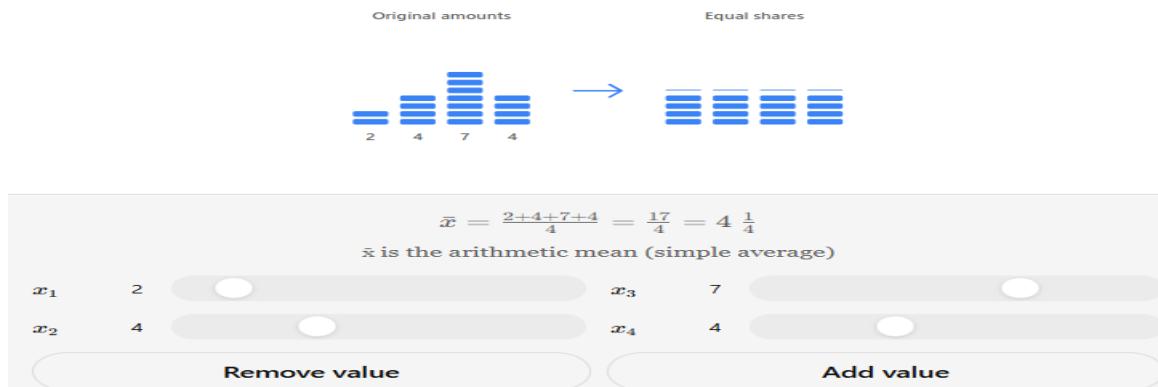
Population Variance: Variance measures how far the values are spread from the mean.

Step 1 – Data

Employee salaries (in lakhs):

4, 5, 5, 6, 7, 8, 8, 9, 30

Step 2 – Mean



Formula:

Mean = Sum of Salaries ÷ Number of Employees

$$= 82 \div 9$$

= ₹9.11 Lakhs

Interpretation: The average salary is ₹9.11 lakhs, but it is increased by the unusually high salary of ₹30 lakhs.

Step 3 – Median

Number of observations = 9

Median Position = $(9 + 1) \div 2 = 5\text{th observation}$

Median = **₹7 Lakhs**

Interpretation: The median represents the middle salary and is not affected by the outlier.

Step 4 – Mode

Salary frequencies:

- ₹5 Lakhs – 2 times
- ₹8 Lakhs – 2 times

Mode = ₹5 Lakhs and ₹8 Lakhs (Bimodal)

Step 5 – Range

Range = Highest Salary – Lowest Salary

$$= 30 - 4$$

= ₹26 Lakhs

Step 6 – Population Variance

Mean = 9.11

Sum of squared deviations = **512.91**

Variance = $512.91 \div 9$

= **56.99**

Interpretation: The high variance is caused by the ₹30 lakh salary.

Step 7 – Population Standard Deviation

Standard Deviation = $\sqrt{56.99}$

= **₹7.55 Lakhs**

This shows that salaries vary by about ₹7.55 lakhs from the average.

Step 8 – Outlier

Outlier: ₹30 Lakhs

Effect:

- Increases the **Mean**
- Increases the **Variance**
- Increases the **Standard Deviation**
- Has little effect on the **Median** and **Mode**

Step 9 – Best Measure of Central Tendency

Best Measure: Median (₹7 Lakhs)

Reason: The ₹30 lakh salary is an outlier that raises the mean. Since most employees earn between ₹4 lakhs and ₹9 lakhs, the median better represents the typical employee salary.

Final Results

Measure	Result
Mean	₹9.11 Lakhs
Median	₹7 Lakhs

Measure	Result
Mode	₹5 Lakhs and ₹8 Lakhs
Range	₹26 Lakhs
Population Variance	56.99
Population Standard Deviation	₹7.55 Lakhs
Outlier	₹30 Lakhs
Best Measure	Median

CASE STUDY 2 – WEBSITE RESPONSE TIME

Problem Statement

A data scientist records the response time (in milliseconds) of a website for **10 requests**.

The response times are:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Using the above data,

1. Calculate the Mean.
2. Calculate the Median.
3. Find the Range.
4. Calculate the Population Variance.
5. Calculate the Population Standard Deviation.
6. Determine whether the data is right-skewed or left-skewed.
7. Explain why the Median is more useful than the Mean.

Given Data

Request	Response Time (ms)
1	120
2	125
3	128
4	130
5	132
6	135
7	140
8	145
9	150
10	300

Number of observations

n = 10

Theory

Mean

The mean is the average response time.

It is obtained by dividing the sum of all response times by the total number of observations.

Median

The median is the middle value after arranging the observations in ascending order.

For an even number of observations,

Median = Average of the two middle values.

Range

The range shows the difference between the highest and lowest response times.

Population Variance

Variance measures how much the response times differ from the average response time.

Population Standard Deviation

Standard deviation is the square root of the variance. It measures the average spread of the response times around the mean.

Step 1 – Data

Website response times (in milliseconds):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Step 2 – Mean

Formula:

Mean = Sum of Response Times ÷ Number of Observations

= 1505 ÷ 10

= **150.5 ms**

Interpretation: The average response time is **150.5 ms**, but it is increased by the unusually high response time of **300 ms**.

Step 3 – Median

There are **10 observations**, so the median is the average of the 5th and 6th values.

$$\text{Median} = (132 + 135) \div 2$$

$$= \mathbf{133.5 \text{ ms}}$$

Interpretation: The median represents the typical response time and is not affected by the outlier.

Step 4 – Range

$$\text{Range} = \text{Highest Response Time} - \text{Lowest Response Time}$$

$$= 300 - 120$$

$$= \mathbf{180 \text{ ms}}$$

Step 5 – Population Variance

$$\text{Mean} = \mathbf{150.5 \text{ ms}}$$

$$\text{Sum of squared deviations} = \mathbf{25581.50}$$

$$\text{Variance} = 25581.50 \div 10$$

$$= \mathbf{2558.15 \text{ ms}^2}$$

Interpretation: The high variance is mainly caused by the **300 ms** response time.

Step 6 – Population Standard Deviation

$$\text{Standard Deviation} = \sqrt{2558.15}$$

$$\approx \mathbf{50.58 \text{ ms}}$$

This indicates that response times vary by about **50.58 ms** from the average.

Step 7 – Skewness

Skewness: Right-Skewed (Positively Skewed)

Reason: Most response times are between **120 ms and 150 ms**, while **300 ms** creates a long right tail and increases the mean.

Step 8 – Best Measure of Central Tendency

Best Measure: Median (133.5 ms)

Reason: The **300 ms** response time is an outlier that raises the mean. The median better represents the normal website response time because it is not influenced by extreme values.

Final Results

Measure	Result
Mean	150.5 ms
Median	133.5 ms
Range	180 ms
Population Variance	2558.15 ms ²
Population Standard Deviation	50.58 ms
Skewness	Right-Skewed (Positively Skewed)
Best Measure	Median

CASE STUDY 3 – STUDENT PERFORMANCE ANALYSIS

Problem Statement

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Using the above data,

1. Find Q1, Q2, and Q3.
2. Calculate the Interquartile Range (IQR).
3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.
4. Calculate the Mean.
5. Calculate the Median.
6. Explain how the outlier affects the Mean.

Given Data

Student	Marks
1	42
2	45
3	48
4	50
5	52
6	52
7	55
8	58
9	60
10	62
11	65
12	95

Number of observations:

$$n = 12$$

Theory

Quartiles

Quartiles divide the data into four equal parts.

- Q1 (First Quartile): 25% of observations lie below Q1.
- Q2 (Second Quartile): Median of the dataset.
- Q3 (Third Quartile): 75% of observations lie below Q3.

Interquartile Range (IQR)

The Interquartile Range measures the spread of the middle 50% of the data.

Formula

$$\text{IQR} = Q3 - Q1$$

Outlier

An outlier is an observation that is much smaller or larger than the other observations.

Outliers are identified using:

$$\text{Lower Limit} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Limit} = Q3 + 1.5 \times \text{IQR}$$

Any value outside these limits is an outlier.

Step 1 – Data

Student marks (ascending order):

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Step 2 – Quartiles

There are **12 observations**.

- **Q2 (Median)** = $(52 + 55) \div 2 = 53.5$
- **Q1** = $(48 + 50) \div 2 = 49$
- **Q3** = $(60 + 62) \div 2 = 61$

Step 3 – Interquartile Range (IQR)

Formula:

$$\text{IQR} = Q3 - Q1$$

$$= 61 - 49$$

$$= 12$$

Step 4 – Outlier Detection

$$\text{Lower Limit} = Q1 - 1.5(\text{IQR})$$

$$= 49 - 18 = \mathbf{31}$$

$$\text{Upper Limit} = Q3 + 1.5(\text{IQR})$$

$$= 61 + 18 = \mathbf{79}$$

Any value below **31** or above **79** is an outlier.

Outlier = 95

Step 5 – Mean

$$\text{Mean} = \text{Sum of Marks} \div \text{Number of Students}$$

$$= 684 \div 12$$

$$= \mathbf{57}$$

Step 6 – Median

Already calculated:

Median = 53.5

Since the mean (**57**) is higher than the median (**53.5**), the high score of **95** increases the average.

Interpretation

The middle 50% of students scored between **49 and 61** marks ($\text{IQR} = 12$), indicating a moderate spread. The score of **95** is an outlier that raises the mean but has little effect on the median. Therefore, the **median** better represents the typical student performance.

Final Results

Measure	Result
Q1	49
Q2 (Median)	53.5
Q3	61
IQR	12
Lower Limit	31
Upper Limit	79
Outlier	95
Mean	57

Measure	Result
Median	53.5

CASE STUDY 4 – E-COMMERCE ORDERS

Problem Statement

An online store records the number of orders received over **10 days** as follows:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Using the above data,

1. Calculate the Mean.
2. Calculate the Median.
3. Find the Mode.
4. Calculate the Population Variance.
5. Calculate the Population Standard Deviation.
6. Calculate the Coefficient of Variation (CV).
7. Explain what the Standard Deviation tells the company.
8. Determine whether the order data is highly or moderately variable.

Given Data:

Day	Number of Orders
1	18
2	20
3	22
4	22
5	24
6	25
7	26
8	28
9	30
10	35

Number of observations:

n = 10

Theory

Mean

The mean is the average number of orders received per day.

Median

The median is the middle value after arranging the data in ascending order.

Mode

The mode is the value that appears most frequently.

Population Variance

Variance measures how widely the daily order counts are spread around the mean.

Population Standard Deviation

Standard deviation indicates the average distance of each day's orders from the mean.

Coefficient of Variation (CV)

The coefficient of variation compares the standard deviation with the mean and expresses variability as a percentage.

Formula

$$CV = (\text{Standard Deviation} \div \text{Mean}) \times 100\%$$

Step 1 – Data

Daily customer orders (ascending order):

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Step 2 – Mean

Formula:

$$\text{Mean} = \text{Sum of Orders} \div \text{Number of Days}$$

$$= 250 \div 10$$

$$= \mathbf{25 \text{ Orders}}$$

Interpretation: On average, the company receives **25 customer orders per day**.

Step 3 – Median

There are **10 observations**.

$$\text{Median} = (24 + 25) \div 2$$

$$= \mathbf{24.5 \text{ Orders}}$$

Step 4 – Mode

The value **22** occurs twice, while all other values occur once.

$$\mathbf{\text{Mode} = 22 \text{ Orders}}$$

Step 5 – Population Variance

$$\text{Mean} = \mathbf{25}$$

$$\text{Sum of squared deviations} = \mathbf{228}$$

$$\text{Variance} = 228 \div 10$$

$$= \mathbf{22.8}$$

Step 6 – Population Standard Deviation

$$\text{Standard Deviation} = \sqrt{22.8}$$

$$\approx \mathbf{4.77 \text{ Orders}}$$

This indicates that daily orders vary by about **4.77 orders** from the average.

Step 7 – Coefficient of Variation (CV)

$$\text{CV} = (\text{Standard Deviation} \div \text{Mean}) \times 100$$

$$= (4.77 \div 25) \times 100$$

$$= \mathbf{19.08\%}$$

Interpretation

The company receives an average of **25 orders per day**, with most values close to the average. The standard deviation and coefficient of variation show that daily sales are fairly consistent, making inventory and staffing easier to plan.

Final Results

Measure

Result

Mean

25 Orders

Measure	Result
Median	24.5 Orders
Mode	22 Orders
Population Variance	22.8
Population Standard Deviation	4.77 Orders
Coefficient of Variation	19.08%
Variability	Moderately Variable

CASE STUDY 5 – COMPARING TWO DATA SCIENCE MODELS

Problem Statement

Two machine learning models have the following prediction errors.

Model A

2, 3, 3, 4, 4, 5, 5, 6

Model B

1, 1, 2, 4, 5, 7, 8, 9

Using the above data,

1. Calculate the mean error for both models.
2. Calculate the population variance.
3. Calculate the population standard deviation.
4. Identify which model has more consistent predictions.
5. Explain whether the model with the lower mean error can always be considered better.

Given Data

Model A	
Observation	Error
1	2
2	3
3	3
4	4
5	4
6	5
7	5
8	6

Model B

Observation	Error
1	1
2	1
3	2
4	4
5	5
6	7
7	8
8	9

Number of observations

n = 8

Theory

Mean Error

The mean error represents the average prediction error made by a model.

Population Variance

Variance measures how far the prediction errors are spread around the mean.

Population Standard Deviation

Standard deviation is the square root of the variance. It indicates how consistent the prediction errors are.

A **smaller standard deviation** means more consistent predictions.

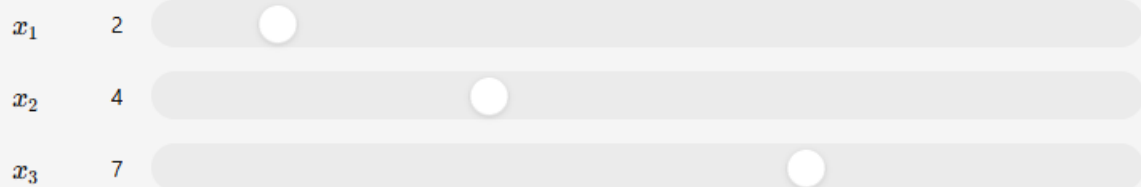
Part A – Model A

Step 1 – Mean



$$\bar{x} = \frac{2+4+7}{3} = \frac{13}{3} = 4 \frac{1}{3}$$

$\bar{x}$ is the arithmetic mean (simple average)



Errors: 2, 3, 3, 4, 4, 5, 5, 6

Mean = Sum of Errors ÷ Number of Observations

$$= 32 \div 8$$

$$= \mathbf{4.00}$$

Step 2 – Population Variance

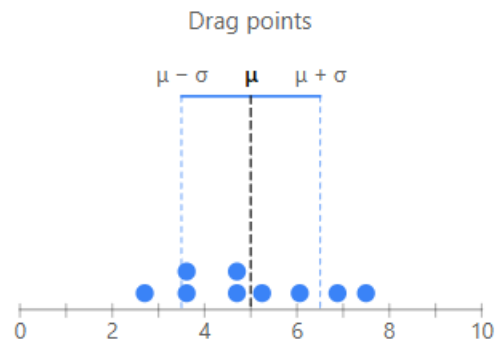
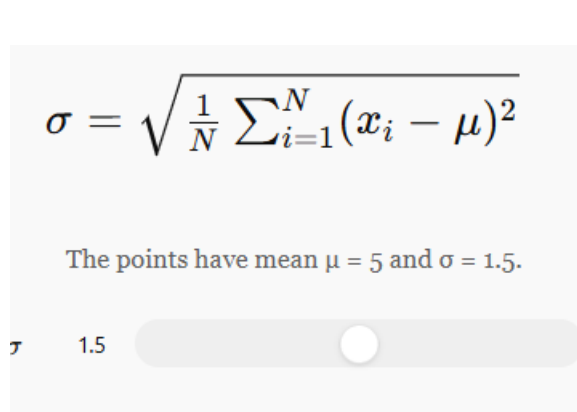
Mean = 4

Sum of squared deviations = 12

$$\text{Variance} = 12 \div 8$$

$$= \mathbf{1.50}$$

Step 3 – Population Standard Deviation



Standard Deviation = $\sqrt{1.50} \approx \mathbf{1.22}$

Part B – Model B

Step 1 – Mean

Errors: **1, 1, 2, 4, 5, 7, 8, 9**

Mean = $37 \div 8$

= **4.625**

Step 2 – Population Variance

Mean = **4.625**

Sum of squared deviations = **69.875**

Variance = $69.875 \div 8$

= **8.734**

Step 3 – Population Standard Deviation

Standard Deviation = $\sqrt{8.734} \approx \mathbf{2.96}$

Comparison of Models

Measure	Model A	Model B
Mean Error	4.00	4.625
Population Variance	1.50	8.734
Population Standard Deviation	1.22	2.96

Which Model is More Consistent?

Model A is more consistent because it has a much lower **variance (1.50)** and **standard deviation (1.22)** than Model B. Its prediction errors remain closer to the average, making its performance more reliable.

Is a Lower Mean Error Always Better?

No. A lower mean error alone does not guarantee a better model. A good machine learning model should have:

- Low mean error
- Low variance
- Low standard deviation
- Consistent performance

Therefore, model evaluation should consider multiple statistical measures rather than only the mean error.

Final Results

Measure	Result
Mean Error (Model A)	4.00
Mean Error (Model B)	4.625
Population Variance (Model A)	1.50
Population Variance (Model B)	8.734
Population Standard Deviation (Model A)	1.22
Population Standard Deviation (Model B)	2.96
More Consistent Model	Model A
Is lower mean error always better?	No, consistency and other evaluation metrics should also be considered.

OVERALL RESULTS AND FINDINGS

Case Study	Dataset	Key Results	Final Finding
Case Study 1 – Employee Salaries	4, 5, 5, 6, 7, 8, 8, 9, 30	Mean = 9.11, Median = 7, Mode = 5 & 8, Range = 26, Variance = 56.99, Standard Deviation = 7.55, Outlier = 30	The salary ₹30 lakhs is an outlier that increases the mean. The median (₹7 lakhs) is the best measure of the typical employee salary because it is not affected by the outlier.
Case Study 2 – Website Response Time	120, 125, 128, 130, 132, 135, 140, 145, 150, 300	Mean = 150.5 ms, Median = 133.5 ms, Range = 180 ms, Variance = 2558.15, Standard Deviation = 50.58 ms, Skewness = Right-Skewed	The 300 ms response time is an outlier that makes the distribution right-skewed. The median better represents the normal website response time.
Case Study 3 – Student Performance	42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95	Q1 = 49, Q2 = 53.5, Q3 = 61, IQR = 12, Outlier = 95, Mean = 57, Median = 53.5	The mark 95 is an outlier identified using the $1.5 \times \text{IQR}$ rule. It increases the mean, while the median remains a better measure of typical student performance.
Case Study 4 – E-Commerce Orders	18, 20, 22, 22, 24, 25, 26, 28, 30, 35	Mean = 25, Median = 24.5, Mode = 22, Variance = 22.8, Standard Deviation = 4.77, Coefficient of Variation = 19.08%	The online store receives 25 orders per day on average. The moderate standard deviation and coefficient of variation indicate stable and moderately variable customer demand.
Case Study 5 – Comparing Two Data Science Models	Model A: 2, 3, 3, 4, 4, 5, 5, 6 Model B: 1, 1, 2, 4, 5, 7, 8, 9	Model A: Mean = 4.00, Variance = 1.50, SD = 1.22. Model B: Mean = 4.625, Variance = 8.734, SD = 2.96	Model A provides more consistent predictions because it has the lowest variance and standard deviation. A lower mean error alone is not sufficient to determine the best model.